

Final Project Reflection: Fast Food Unhealthiness and Popularity

For this project, I decided to explore the relationship between fast-food menu unhealthiness and overall restaurant popularity. The main research question I investigated was: *Does higher unhealthiness relate to higher popularity?* I was interested in this topic because I like fast food (specifically Chick-fil-A) and it's a major part of everyday life, and many people tend to choose restaurants based on convenience and popularity rather than its nutritional value. This led me to question if there is a correlation between how unhealthy a menu is and how successful a chain becomes.

To conduct my analysis, I used two datasets from Kaggle. From the nutrition dataset, I specifically selected key variables (calories, sugar, saturated fat, trans fat, and sodium) to represent unhealthiness. From the second dataset, I focused on systemwide sales, average sales per unit, and total store count to measure popularity. Combining these datasets allowed me to directly compare nutritional quality with restaurant success.

Specifically for my analysis, I limited my focus to major fast-food chains such as McDonald's, Burger King, Subway, Taco Bell, Chick-fil-A, Sonic Drive-In, Dairy Queen, and Arby's, as these were the only chains with complete nutritional data available.

Since there is no set score to determine what food is considered healthy or not, I created a custom health score inspired by the Nutrient Profile Model (NPM) calculator. This score combined sugar, saturated fat, trans fat, and sodium into a single value to represent how unhealthy a food item is. Each nutrient was adjusted based on total calories to allow fair comparisons across items of different sizes. I then compared each nutrient to benchmark thresholds, which helped standardize the score and reflect how much each item exceeded recommended limits. The final score represented overall nutritional risk, where higher value indicates an unhealthier diet.

Using this score, I calculated average unhealthy scores for each restaurant and compared them to measures of popularity such as systemwide sales and total store count. I created several visualizations, including a scatterplot comparing unhealthy scores to revenue, a heatmap of average nutrition values, and bar charts showing store count and sales comparisons.

The results showed that there is not a perfect direct relationship between unhealthiness and popularity. While many high-selling chains did have moderately high unhealthy scores, the most popular restaurant, McDonald's, did not have the highest unhealthy score. This suggests that although unhealthy foods may contribute to popularity, other factors such as brand recognition, pricing, and accessibility play a significant role in driving sales.

Based on these findings, my overall argument is that unhealthiness alone does not determine popularity, but it can be one contributing factor among many. Throughout this project, I learned how to combine multiple datasets and create a meaningful metric to analyze a complex real

world question. I also gained experience in cleaning data, normalizing values, and creating visualizations that clearly communicate results.

One thing that went well was the development of the health score, which provided a structured way to compare different restaurants. Additionally, the visualizations helped make the patterns in the data easier to understand. However, there were also some limitations and challenges. Different restaurants had different numbers of menu items, which could have affected the stability of the averages. Additionally, popularity was only measured using sales and store count, which does not include factors such as customer preferences, advertising, or pricing. Another challenge was ensuring that the health score accurately reflected nutritional quality without oversimplifying it.

Overall, this project strengthened my understanding of how data can be used to analyze realistic behaviors and trends. It highlighted that meaningful analysis requires considering multiple factors rather than relying on a single metric. This experience improved my ability to think critically and clearly communicate insights.

Citations

1. Fast Food Nutrition Dataset

Pedersen, U. T. (n.d.). *Fast food nutrition*. Kaggle.

<https://www.kaggle.com/datasets/ulrikthygepedersen/fastfood-nutrition>

2. Top 50 Fast-Food Chains Dataset

Done, S. (n.d.). *Top 50 fast-food chains in the U.S.* Kaggle.

<https://www.kaggle.com/datasets/stetsondone/top50fastfood>

3. Nutrient Profile Model (NPM) Calculator

Global Health Advocacy Incubator. (n.d.). *Reducing demand for unhealthy food: NPM calculator*.

<https://www.advocacyincubator.org/program-areas/food-and-nutrition/reducing-demand-for-unhealthy-food/npm-calculator>